15.5 Experimental and Theoretical Probability

Lesson Introduction

Benchmarks	MA.7.MA.DP.2.3 Find the theoretical probability of an event related to a simple experiment. MA.7.MA.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.	Learning Targets	• I can perform a simulation to determine an experimental probability. • I can find the theoretical probability of an event. • I can compare the experimental probability from a simulation to the theoretical probability of a simple experiment.

Benchmark Coherence	Building On	Working Toward
	N/A	MA.8.DP.2.2 Find the theoretical probability of an event related to a repeated experiment. MA.8.DP.2.3 Solve real-world problems involving probabilities related to single or repeated experiments, including making predictions based on theoretical probability.

Essential Questions	• What is the difference between theoretical and experimental probability? • How do you determine the theoretical and experimental probabilities of a simple experiment?
Lesson Narrative	In this lesson students are introduced to the terms “theoretical probability” and “experimental probability” and given opportunities to explore the difference. Students compare the theoretical and experimental probabilities of a simple experiment over different numbers of trials to build a strong foundational understanding of the difference between the two concepts. Students also begin exploring how the number of trials impacts the closeness of the experimental probability to the theoretical probability, a concept that will be further developed in Lesson 6.

Component Summary				
	15.5.1 Warm-Up (~5 min)	Determine equivalent probabilities (SE pg. 421)	15.5.4 Guided Instruction (5 min)	Introduce theoretical and experimental probability (SE pg. 425)
	15.5.2 Exploration (15-20 min)	Simulate a simple experiment (SE pg. 421)	15.5.5 Try It (5-10 min)	Practice interpreting theoretical and experimental probability in context (SE pg. 425)
	15.5.3 Bring It Together (10 min)	Observe patterns after simulating a simple experiment (SE pg. 423)	15.5.6 Wrap-Up (~5 min)	Summarize the difference between theoretical and experimental probability (SE pg. 426)

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • Theoretical probability • Experimental probability <u>Additional Terms:</u> • Probability	• 6-sided number cube • Calculators	• This is a vocabulary-rich lesson. Consider preparing your classroom-specific vocabulary or glossary strategies for support (Word Wall, Math Cards, etc.)

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may confuse theoretical and experimental probabilities. • Students may make mistakes when converting from a fraction to a decimal.	• Consider having students summarize the definitions of theoretical and experimental probability in their own words through a Turn-and-Talk during the 15.5.4 Guided Instruction.
	Literacy and Language Routines	**Mathematical Thinking and Reasoning (MTR) Standard**
	Notice and Wonder In Exploration 15.5.2 have students engage in a Notice and Wonder. What do you notice about the points on the graph? What do you wonder? As the number of rolls increases, what do you notice happening on the graph?	**MA.K12.MTR.4.1: Discussion and Reflection** This lesson creates many opportunities for students to discuss their thinking with peers. In 15.5.3 Bring It Together, students have a chance to engage in discourse as a class regarding theoretical vs. experimental probability.
Differentiate Instruction	In the 15.5.4 Guided Instruction, the definitions of theoretical and experimental probability are wordy. Have students go through and highlight or underline important words. Students can each come up with a definition for theoretical and experimental probability in their own words.	

Lesson Notes:

15.5.1 Warm-Up: Marbles

Component Overview: *Students complete two statements that represent equivalent probabilities in the event of drawing a red marble from a bag.*

15.5.1 Warm-Up: Marbles

1. Fill in the blanks in the statements below so the probability of drawing a red marble from either bag is the same. Use any number 1 through 9 at most one time each.
   Answers will vary.

   There are 2 red marbles and 4 blue marbles in Bag A.

   There are 3 red marbles and 6 green marbles in Bag B.

- *How can you show that the probabilities are equivalent?*
- *What is the probability expressed as a percentage?*

15.5.2 Exploration: Playing the Long Game

Component Overview: *In this Exploration, students explore experimental probability for the first time. They start by coming up with the theoretical probability first and then complete the experiment while recording their results. Students compare the theoretical and experimental probabilities.*

15.5.2 Exploration: Playing the Long Game

Samaria plays a game with a fair six-sided die in which she only wins if she rolls a 1 or a 2.

1. List the outcomes in the sample space for rolling the fair die.

   $1, 2, 3, 4, 5,$ *or* 6

2. Discuss the probability of Samaria winning the game with your partner or group. Then, complete the statement.

   The probability Samaria wins the game, represented as a fraction, is $\frac{2}{6}$ *or* $\frac{1}{3}$ because there are 2 possible winning outcomes out of 6 total possible outcomes from rolling the die.

3. If Samaria is given the option to flip a coin and win if it comes up heads, explain whether or not that would be a better option for her to win using probability.

   This would be a better option because the probability with the dice is $\frac{2}{6}$ *or* 33.3% *and the probability for a coin flip landing heads up is* $\frac{1}{2}$ *or* 50%

This Exploration can be done in partners or groups of 3-4. Data is gathered from each group and used in the 15.5.3 Bring It Together, so having fewer groups may make gathering the data easier.

4. With your partner or group, follow these instructions 10 times to create the graph.
   - One person rolls the number cube. Everyone records the outcome.
   - Calculate the fraction of all rolls that are wins for Samaria so far. Approximate the fraction with a decimal value rounded to the hundredths place using a calculator. Record both the fraction and the decimal in the last column of the table.
   - On the graph, plot the number of rolls and the fraction of rolls that were wins.
   - Pass the number cube to the next person in the group.

Answers will vary.

Roll	Outcome	Total Number of Wins for Samaria	Fraction of Games Played that are Wins
1	3	0	$\frac{0}{1} = 0.0$
2	2	1	$\frac{1}{2} = 0.5$
3	3	1	$\frac{1}{3} = 0.33$
4	4	1	$\frac{1}{4} = 0.25$
5	5	1	$\frac{1}{5} = 0.2$
6	1	2	$\frac{2}{6} = 0.33$
7	1	3	$\frac{3}{7} = 0.43$
8	3	3	$\frac{3}{8} = 0.38$
9	5	3	$\frac{3}{9} = 0.33$
10	4	3	$\frac{3}{10} = 0.3$

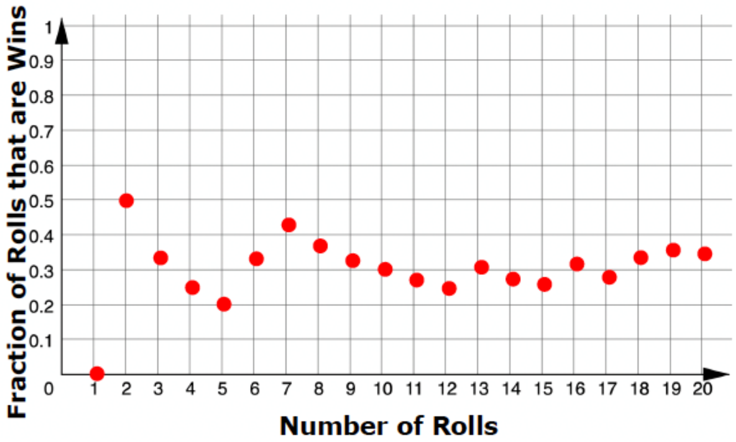
What is the likelihood of Samaria winning the game?

If you do not have a number cube, cut pieces of paper labeled 1-6 and randomly choose with replacement.

Consider going over how to find the fraction of games played that are wins. Consider posting this on the white board to help.

$$\frac{\text{number of wins}}{\text{number of trials}}$$

15.5.2 Exploration: Playing the Long Game (continued)



5. With your partner or group, discuss what you notice is happening with the points on the graph.
   Answers will vary.
   a. After 10 rolls, what fraction of the total rolls were a win?
      Answers will vary. In the example shown the fraction of rolls that were a win is $\frac{3}{10}$.

   b. How close is this fraction to the probability that Samaria will win?
      Answers will vary.

Notice and Wonder

- *What do you notice about the points on the graph?*
- *What do you wonder?*
- *As the number of rolls increases, what do you notice happening on the graph?*

6. Roll the number cube 10 more times. Record your results in this table and add them to the graph from earlier.
Answers will vary.

Roll	Outcome	Total Number of Wins for Samaria	Fraction of Games Played that are Wins
11	6	3	$\frac{3}{11} = 0.27$
12	6	3	$\frac{3}{12} = 0.25$
13	2	4	$\frac{4}{13} = 0.31$
14	3	4	$\frac{4}{14} = 0.29$
15	3	4	$\frac{4}{15} = 0.27$
16	1	5	$\frac{5}{16} = 0.31$
17	3	5	$\frac{5}{17} = 0.29$
18	1	6	$\frac{6}{18} = 0.33$
19	1	7	$\frac{7}{19} = 0.37$
20	4	7	$\frac{7}{20} = 0.35$

7. Observe any trends on the graph after 20 rolls.

   a. After 20 rolls, what fraction of the total rolls was a win?
   Answers will vary. In the example shown $\frac{7}{20}$ rolls resulted in a win.

   b. How close is this fraction to the probability that Samaria will win?
   Answers will vary. It is close to the probability of Samaria winning, $\frac{2}{6}$, or $\frac{1}{3}$.

Notice and Wonder

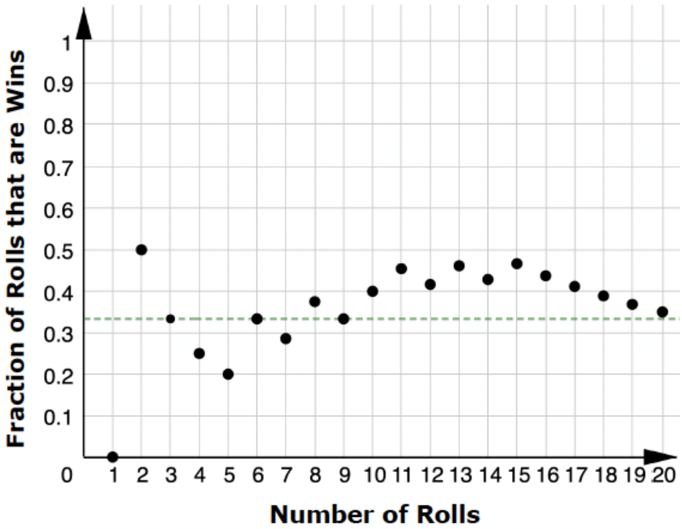
- *What do you notice about the table?*
- *What do you wonder?*
- *As the number of rolls increases, what do you notice happening in the table?*
- *How does this relate to what is happening in the graph?*

15.5.3 Bring It Together: Playing the Long Game

Component Overview: *As a class, students will look at the data they have collected and compare it to the probability that Samaria would win the game.*

15.5.3 Bring It Together: Playing the Long Game

A sample graph from a group who performed the same simple experiment in the previous activity is shown.



1. What does the green dotted line represent on the graph?
   The dotted line represents the probability of rolling a 1 or a 2, $\frac{2}{6}$, or $\frac{1}{3}$

Students may think the dotted line represents an average probability, rather than the probability of winning.

15.5.3 Bring It Together: Playing the Long Game (continued)

2. Consider the data from all of the groups performing the experiment.

   a. Complete the table below with your class by adding the overall data from each group. The first row has already been completed using the sample data above.
   Answers will vary.

Group	Total Number of Wins for Samaria	Fraction of Games Played that are Wins
Sample Data	7	$\frac{7}{20} = 0.35$

   b. Write the fraction that represents the total number of wins out of the total number of games played by all groups.
   Answers will vary.

   c. How close is this fraction to the probability that Samaria will win?
   Answers will vary.

- *Do you think the number of trials impacts how close the total number of wins out of the total number of games played is to the predicted probability of winning?*
- *Do you think that if you conducted this experiment again, you would get the exact same results?*

The probability for an event represents the proportion of the time it is expected for the event to occur in the long run, over many trials. It does not mean that is exactly how many times the event will actually occur.

15.5.4 Guided Instruction: Experimental and Theoretical Probability

Component Overview: *Students are introduced to the definitions of theoretical and experimental probability before summarizing the Exploration activity using these terms.*

15.5.4 Guided Instruction: Experimental and Theoretical Probability

Consider the terms defined below and how they relate to the previous Exploration.

A ***theoretical probability*** is a number between 0 and 1 that represents the likelihood of an event in a theoretical model based on a sample space. If all outcomes in the sample space are equally likely, then the theoretical probability of an event is the ratio of the number of outcomes in the event to the number of outcomes in the sample space.

An ***experimental probability*** is a number between 0 and 1 that shows the ratio of the number of times an event occurs to the total number of trials or times the activity is performed.

Highlight or underline important words in the definition. Consider giving students the opportunity to turn to a partner and share what they understand about each term in their own words.

1. Use the definitions to complete the statements.

In the Exploration, the [○ experimental / ○ theoretical] probability of Samaria winning the game was $\frac{1}{3}$ because there were [○ 2 / ○ 3 / ○ 6] possible outcomes out of [○ 2 / ○ 3 / ○ 6] in the sample space that won. By perfoming the activity, each group determined [○ an experimental / ○ a theoretical] probability for the experiment.

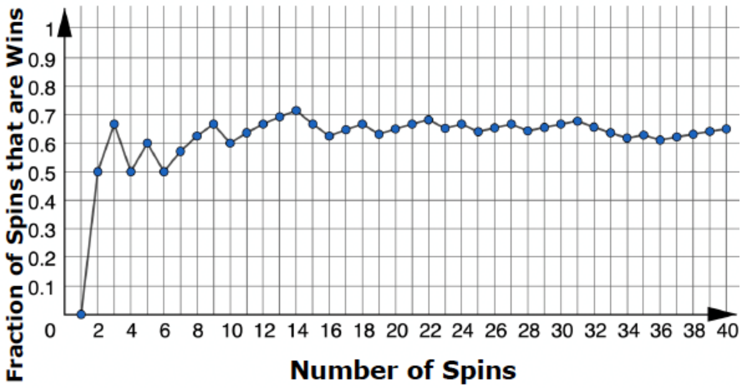
15.5.5 Try It: Experimental and Theoretical Probability

Component Overview: *Students apply what they have learned about experimental and theoretical probability. Students should be able to determine the difference between which probabilities are being found.*

15.5.5 Try It: Experimental and Theoretical Probability

1. Deepa wants to know if flipping a quarter really does have a probability of $\frac{1}{2}$ of landing heads-up, so she flips a quarter 10 times. It lands heads-up 3 times and tails-up 7 times. Explain whether or not she has proven that the probability of landing heads up is not $\frac{1}{2}$.
Deepa has found the experimental probability, this may not be exactly $\frac{1}{2}$. If she continues the trials over and over, the probability average will get closer and closer to $\frac{1}{2}$.

2. A spinner is spun 40 times for a game. The graph shows the fraction of games that are wins.



   a. Estimate the probability of a spin winning this game based on the graph.
   Answers will vary but should be around 0.6 *or* 60%

   b. What type of probability is your estimate?
   Experimental Probability.

Consider asking students the following:

- *How many trials did Deepa conduct?*
- *Do you think that was enough trials?*
- *Do you think an experimental probability being different from the theoretical probability means that the theoretical probability was wrong?*

15.5.6 Wrap-Up: Cool Down

Component Overview: *Students summarize what they have learned today with a quick note to a friend.*

15.5.6 Wrap-Up: Cool Down

1. Your friend was absent today and will need the notes from class. Write a quick note to help them understand the difference between the theoretical and experimental probability of a simple experiment.
   Answers will vary.

Consider using data gathered from this Wrap-Up to inform differentiation strategies in Lessons 6 and 7 as students will be applying experimental and theoretical probability in the following lessons.